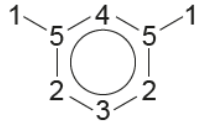


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		5 environments correctly identified from the ^{13}C NMR spectrum and indicated on drawn structure of 1,3-dimethylbenzene (1) signals at 126-136 δ identified as aromatic (1) signal at 20 δ identified as aliphatic / CH_3 (1) 	1		2	3		
		(v)		must be an aromatic compound so at least 6 carbon atoms (1) molecular formula therefore likely to be $\text{C}_8\text{H}_6\text{O}_2$ (1) there are 2 additional carbon atoms therefore likely to be a dialdehyde or cannot be acid-aldehyde or alcohol-aldehyde or acid-alcohol as molecular formula does not fit (1) made by partial oxidation (of each methyl group) (1) formula is  (1)		2	3	5		